

ENTERPRISE MULTI-TENANT SAAS PLATFORM

Project Overview

Company management platform streamlining corporate HR operations, employee attendance, automated payroll processing, CRM, and role-based operational dashboards.

Problem Statement:	SMEs rely on fragmented spreadsheets leading to attendance discrepancies, payroll errors, and zero real-time operational visibility.
Solution Implemented:	Architected a unified multi-tenant SaaS application with granular role-based permissions (Super Admin, HR, Manager) and automated payroll engines.
My Technical Role:	Full-Stack SaaS Architect

Main Features & Technical Stack

Core Features:	Technologies Used:
- Role-based dashboards (Super Admin, HR Specialist, Manager) - Automated attendance tracking and leave approval workflows - Dynamic salary receipt breakdown & payroll calculator - Client CRM pipeline and system audit logs	Next.js, React, TypeScript, Tailwind CSS, Cloudflare Workers, Cloudflare D1, Drizzle ORM, Vercel
	Architecture Flow:
	Next.js App Router → Serverless API Routes → RBAC Middleware → Drizzle ORM → Cloudflare D1 SQL DB

Technical Challenge & Results

Key Engineering Challenge: Enforced server-side middleware authorization to prevent unauthorized access across administrative routes.
Validated Results: Delivered responsive enterprise ERP handling multi-department operations with sub-100ms API response times.